Controlled sub-nanometer tuning of photonic crystal resonator by carbonaceous nano-dots

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Abstract: We propose and demonstrate a scheme that enables spectral tuning of a photonic crystal high-quality resonant mode, in steps finer than 0.2 nm, via electron beam induced deposition of carbonaceous nano-dots. The position and size of a nano-dot with a diameter of <100 nm are controlled to an accuracy on the order of nanometers. The possibility of selective modal tuning is also demonstrated by placing nano-dots at locations pre-determined by theoretical computation. The lasing threshold of a photonic crystal mode tends to increase when a nano-dot is grown at the point of strong electric field, showing the absorptive nature of the nano-dot.

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1. Introduction

A single-cell photonic crystal (PhC) cavity supports high-quality (Q) resonant modes confined in a cubic wavelength region [1-3]. Light emission from an active material can efficiently be coupled to these modes and thereby low threshold lasers [2-5], single photon sources [6-9], and cavity quantum electrodynamics (cavity-QED) effects [9-14] can be demonstrated. In these demonstrations, the precise spectral matching between the PhC cavity mode and atom-like emitter is particularly required to maximize the light-matter interaction. However, the resonance of the PhC cavity is generally detuned from a desired wavelength due to unavoidable crystal growth and fabrication imperfections. In addition, it is necessary to engineer an optical resonator independently in on-chip optical integrated systems as well [1, 15]. To tune a resonant frequency and overcome the spectral mismatching, post-processing tuning methods, such as wet chemical digital etching [16], atomic force microscope (AFM) nano-oxidation [17], and photosensitive tuning of chalcogenide glasses [18-19], have been suggested. However, due to the ultra-small size of the PhC cavity, it has not been achieved to control optical properties of selected resonant modes reliably and independently without sensitive environmental influences such as humidity in the AFM oxidation process. Therefore, a predictable and selective tuning of a desired mode in a high-Q, ultra-small PhC cavity still remains a challenge.

In general, four different resonant modes appear in a triangular-lattice single-cell PhC cavity: monopole, quadrupole, hexapole and dipole modes [20, 21]. One can selectively modify the optical properties of a desired resonant mode, *e.g.* a resonant frequency, due to the completely distinguishable electric field intensity profile of each mode. If a nanostructure is introduced in the intensity antinode of a mode, optical properties of that mode would be severely altered. On the other hand, a nanostructure placed in the intensity node cannot affect the optical properties. In this letter, we propose a novel method for selective and repeatable spectral tuning of a PhC resonant mode based on the deposition of carbonaceous nano-dots (CNDs) in a location pre-determined by theoretical computation.

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2. Electron beam induced deposition of carbonaceous nano-dots

Electron beam induced deposition (EBID) using a scanning electron microscope (SEM) is one of the most promising methods to fabricate nanostructures such as nano-tips and thin films at predetermined positions [22-25]. This makes it particularly interesting for applications such as selective tuning of a desired mode in a PhC cavity. In the EBID, proper precursors injected into the chamber of the SEM are trapped by the highly focused electron beam and deposited on the surface of a substrate as shown in Fig. 1(a). Using this method, one can fabricate complicated 3-D nanostructures composed of various materials, *e.g.* metals or semiconductors [26, 27]. Even if there is no injection of a precursor, carbonaceous nanostructures can be formed in an oil-type SEM. In this case, organic molecules of diffusion vacuum pump oil used in the SEM are deposited on the substrate. For an efficient tuning, we deposit these extremely small solid CNDs in a desired position of a high-Q PhC cavity.

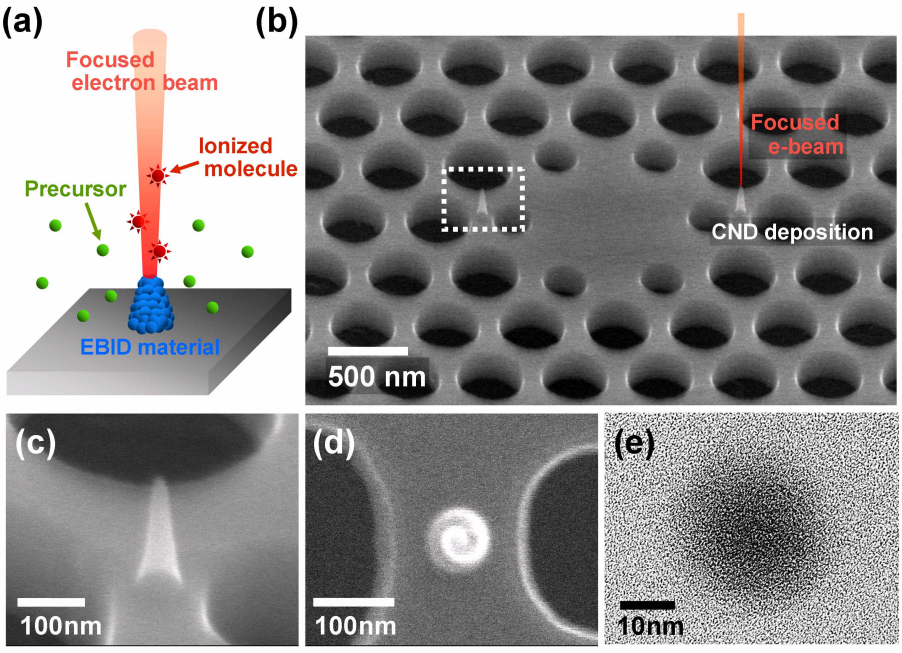


Fig. 1. Generation of carbonaceous nano-dots. (a) Schematic of electron-beam-induced CND deposition mechanism. (b) CNDs formed in a PhC cavity. (c) Magnified image of the dotted region of Fig. 1b. (d) Top view. (e) High resolution TEM image of a CND showing no crystalline signature; the CND is deposited on a TEM grid.

Figure 1(b) shows that two CNDs are accurately located in the small region between two air holes of a fabricated single-cell PhC cavity. CNDs with a typical size of <100 nm are deposited on the surface with negligible fabrication errors. The SEM (Hitachi S-4300, field emission type) with an acceleration voltage of 30 kV and a beam current of 10 pA is used to form these CNDs. The ambient pressure of the SEM chamber is ~7 x 10^{-7} Torr. In the spot mode with 500K magnification, the focused electron beam is shone at the position of choice. Figs. 1(c) and 1(d) show the magnified SEM images (side and top views, respectively) of the CND fabricated in Fig. 1(b). Note that the CND has a spiral-shaped cross section [Fig. 1(d)]. When an electron passes through an electromagnetic lens of the SEM, a force by the vertical component of the magnetic field causes the electron to move along the spiral path. In addition, high-resolution transmission electron microscope (HR-TEM) analysis [Fig. 1(e)] shows that the CND has amorphous material characteristics. The absence of any bright spots in transmission electron diffraction pattern is evidence that the CNDs have an amorphous structure.

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The electron beam focusing and the deposition time are crucial to the shape and size of the CND as shown in Fig. 2. Figure 2(a) exhibits the progressive change of the shape and size of CNDs. In the optimal focusing condition of the SEM, the smallest and spiral-shaped CND is obviously fabricated (the rightmost one). In addition, longer deposition time causes larger CNDs as shown in Fig. 2(b). This controllable shape and size of the CND will provide new opportunities to engineer photonic crystals with various optical properties.

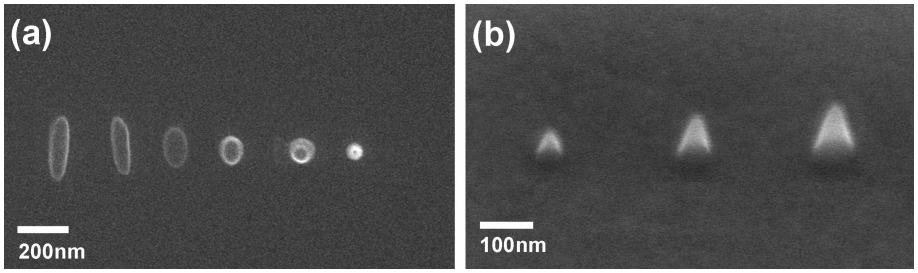


Fig. 2. Tuning of SEM for optimum focusing. (a) The progressive change of the shape and size of CNDs. Left to right, the electron beam becomes in the optimal focusing condition. (b) The SEM image of three CNDs with deposition time of 60, 90, and 120 seconds (left to right), respectively, taken at an angle of 15 degrees.

3. Controlled sub-nanometer tuning of a photonic crystal resonant mode

First, we fabricate a single-cell PhC cavity using a 200-nm-thick InGaAsP heterostructure slab structure grown by metal organic chemical vapor deposition. An InGaAsP single quantum well (QW), the thin white layer in Fig. 3(a), is embedded in the middle of the slab. The lattice constant (*a*) of the triangular PhC pattern is about 550 nm and the radii of the PhC air holes and the 6 nearest air holes are 0.35 *a* and 0.22 *a*, respectively. We choose these structural parameters to locate the nondegenerate hexapole and two doubly degenerate quadrupole modes, among the four resonant modes excited in the PhC single-cell cavity, within the electronic band gap of the active material that has a central emission peak of 1550 nm and a spectral width of ~100 nm [21]. The electric field intensity profiles of the hexapole and quadrupole modes computed by 3-D FDTD simulation are shown in Fig. 6(a) (right column). The PhC pattern is defined by electron beam lithography and chemically assisted ion beam etching processes. HCl wet-etching process is followed to remove the underlying InP layer. Figure 3(b) shows the SEM image of a fabricated PhC cavity.

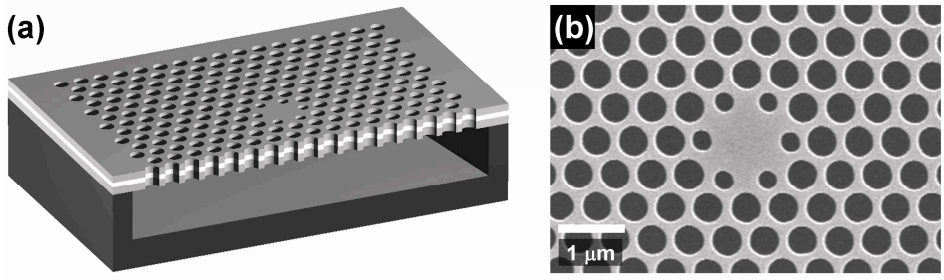


Fig. 3. PhC resonant cavity. (a) Schematic of the PhC slab cavity. (b) SEM image of a fabricated single-cell PhC laser cavity.

The diameter and height of a CND critically depend on the deposition time as we mentioned in the previous section [22-25]. In fact, it is possible to form a CND with an exactly desired size by controlling deposition time and sequential steps of deposition. For example, Fig. 4(a) shows the growth album of a CND with sequentially increasing its diameter. The diameter of the CND was initially ~50 nm (left image) and then the diameters increase up to 70 nm (middle) and 90 nm (right) by one and two more steps of deposition,

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respectively. With each deposition step, the diameter of the nano-dot increases by ~5 nm. One can easily expect that the sub-nanometer tuning of a resonant frequency can be achieved in the PhC cavity by introducing such an extremely small nanostructure with exactly controlled size and position.

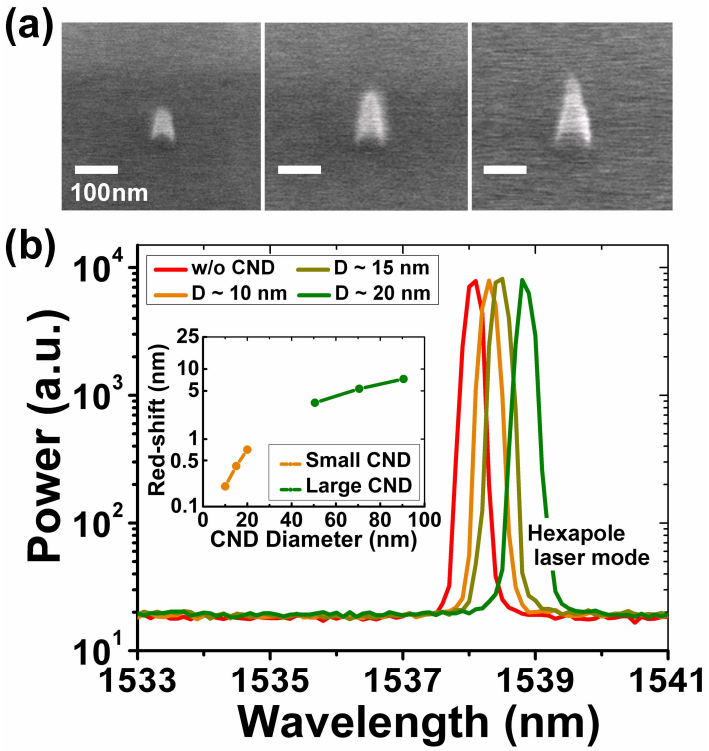


Fig. 4. Sub-nanometer spectral tuning by controlled CND growth. (a) Growth of a CND by sequential re-depositions. The diameter of the CND can be controlled by a step of ~5 nm. (b) Sub-nanometer tuning of a hexapole laser mode. A CND is deposited at the center of the single-cell PhC cavity. Each additional CND deposition red-shifts the hexapole mode by a step of ~0.2 nm.

In order to investigate the resonance tuning induced by the CNDs, photoluminescence (PL) spectroscopy is carried out in the PhC cavities with different size of the CND. The PhC structures are optically pumped with 10-ns-long pulses repeating every 1 μs, using a 980-nm laser diode at room temperature. Then the hexapole mode laser is solely observed. All PL spectra are measured at the incident pump power level of ~300 μW that is about ten times larger than the laser threshold. A small CND with a diameter of <20 nm is fabricated at the center of the PhC cavity as shown in Fig. 7(a) and then the diameter of the CND increases through sequential steps of deposition. As shown in the spectra of Fig. 4(b), by introducing the CND and controlling its size by the step of 5 nm from 10 nm to 20 nm, a precise and continuous resonance red-shift with the spectral resolution of ~0.2 nm is observed. On the other hand, if a CND with a larger diameter of >50 nm is deposited, enhanced red-shift of a few nanometers is observed as shown in the inset of Fig. 4(b). The successful demonstration of the controllable sub-nanometer tuning, which is comparable to the linewidth of the high-Q PhC resonant mode, is a meaningful step for lasers and quantum optics systems. Furthermore, the ultra-small size of the CND, which is only a few tens of nanometer, enables to control and tune a specific resonator or an optical element in a quantum information and computation system [9, 28].

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In addition, it is possible to achieve a stepped tuning through the deposition of additional CNDs in different positions as shown in Fig. 5. One, two, and three CNDs with diameters of ~70 nm are deposited on a resonant cavity [Fig. 5(a)]. As shown in the measured spectra of a single hexapole laser mode and two quadrupole non-lasing modes [Fig. 5(b)], the spectral tuning depends on the number of the CNDs. Each CND generates a red-shift of ~3 nm.

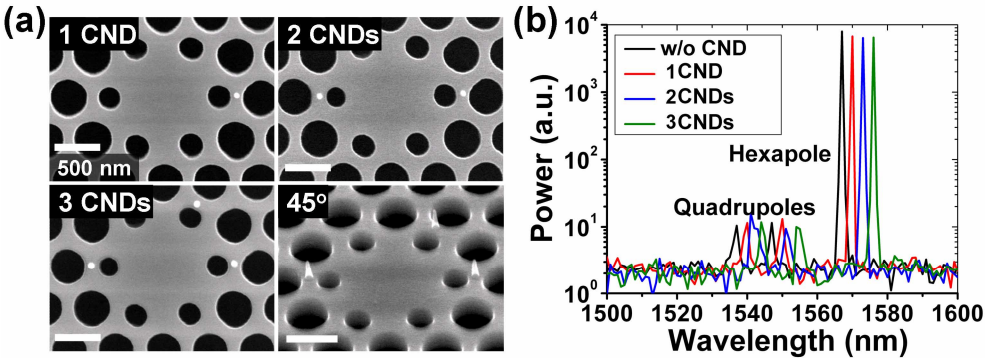


Fig. 5. Spectral tuning by sequential deposition of CNDs. (a) One, two, and three CNDs of diameter ~70 nm on a resonant cavity. (b) Measured spectra of hexapole and quadrupole modes. Each CND generates a red-shift of ~3 nm.

4. Selective tuning of a photonic crystal resonant mode

The possibility of selective modal tuning is demonstrated by placing nano-dots at locations pre-determined by theoretical computation. The optical properties of the PhC resonant modes are modified by the position as well as the size of the CNDs on the cavity. We can place the CNDs in the strong or weak electric field regions of a specific cavity mode to investigate how selectively and effectively the optical properties of the mode are affected by the CNDs. In Fig. 6(a), six CNDs are located around the cavity where the hexapole mode has antinodes of electric field intensity [20, 21]. On the other hand, two doubly degenerated quadrupole modes have relatively weaker electric field intensity in these regions as shown in electric field intensity profiles computed by 3-D finite-difference time-domain (FDTD) simulation (right, Fig. 6(a)). Therefore, this cavity structure with the six CNDs can cause a severe resonance red-shift particularly in the hexapole mode. As shown in Fig. 6(b), 14.0 nm-red-shift of the hexapole lasing mode were measured. Such a red-shift of the hexapole mode is larger than those of the two quadrupole modes, which are 8.1 and 9.0 nm, respectively, as expected. These results imply that selective tuning of a specific resonant mode is successfully demonstrated by taking into account of the field profile of the mode. In addition, the resonance shift depends on not only the local field intensity at the position of the nano-dot but also total overlap of the resonant mode with the nano-dot. In Fig. 6, the local field intensity can be more dominant factor to explain the different resonance shifts of the hexapole and quadrupole modes because they have similar mode volumes [20].

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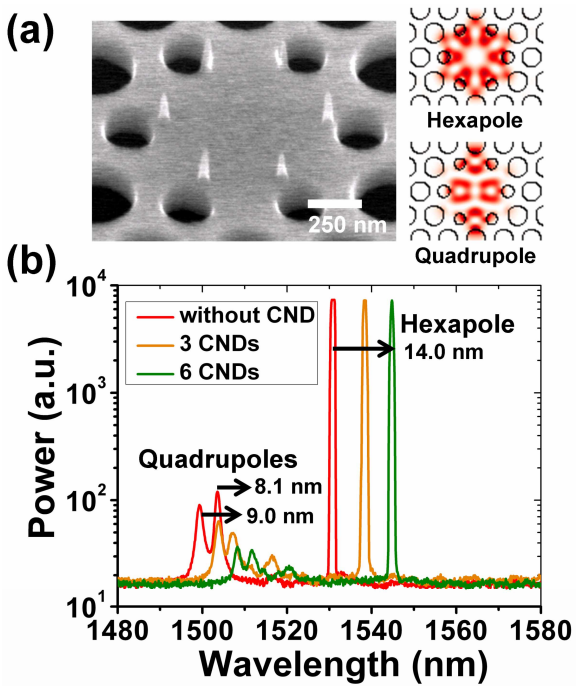


Fig. 6. Selective resonance tuning by CNDs. (a) Six 60-nm CNDs are deposited near the antinodes of a hexapole mode. Electric field intensity profiles of the hexapole and quadrupole modes are shown to the right. (b) A particular placement of CNDs results in a red-shift of the hexapole mode larger than those of the quadrupole modes.

5. Effects of nano-dots on laser thresholds

Further study about the effect of a CND on a resonant mode is achievable by measuring threshold of the laser launched by a resonant mode of interest. We investigate two scenarios: a CND is placed at the center of the PhC cavity [Fig. 7(a)] or at the position close to a nearest-neighbor air-hole around the cavity [Fig. 7(b)]. PL study is performed to measure spectrum and lasing threshold. Then we observe the single hexapole mode lasing operations in both cavities. The electric-field energy density profile of the hexapole mode is computed by FDTD method and superimposed on the SEM images of the PhC single-cell lasers [Figs. 7(a) and 7(b)]. Note that the hexapole mode has an electric field intensity node at the cavity center and an antinode near the first layer of air-holes. The effects of the CND on the lasing operation are obviously measured in Figs. 7(c) and 7(d), where lasing peak intensities are plotted as a function of the incident pump power. Each lasing threshold of the cavity with CND ranges from ~20 μW to ~40 μW and is normalized by threshold of the cavity without CND. When the CND is in the central intensity node of the electric field profile, a small threshold increment of ~25 % is observed [Fig. 7(c)]. On the other hand, the CND located in the strong electric field region of the hexapole mode causes a larger increment of the lasing threshold as much as ~150 % [Fig. 7(d)]. This observation shows that the position of a CND strongly affects the optical properties of a desired mode. In addition, the resonance shifts by the CNDs of Figs. 7(a) and 7(b) are ~1.6 nm and ~3.4 nm, respectively.

In order to estimate the Q factor of the laser mode, we estimated the full-width at half-maximum (FWHM) of the lasing peak at a transparency pumping level, which is ~80 % of the threshold pump power. The FWHM is limited by the spectral resolution of our monochromator (~0.3 nm), even when the CND is deposited on the cavity. The FWHM of <~0.3 nm corresponds to the cavity Q factor of >~5000 that is still high enough to do measurements of quantum optical phenomena [6, 9].

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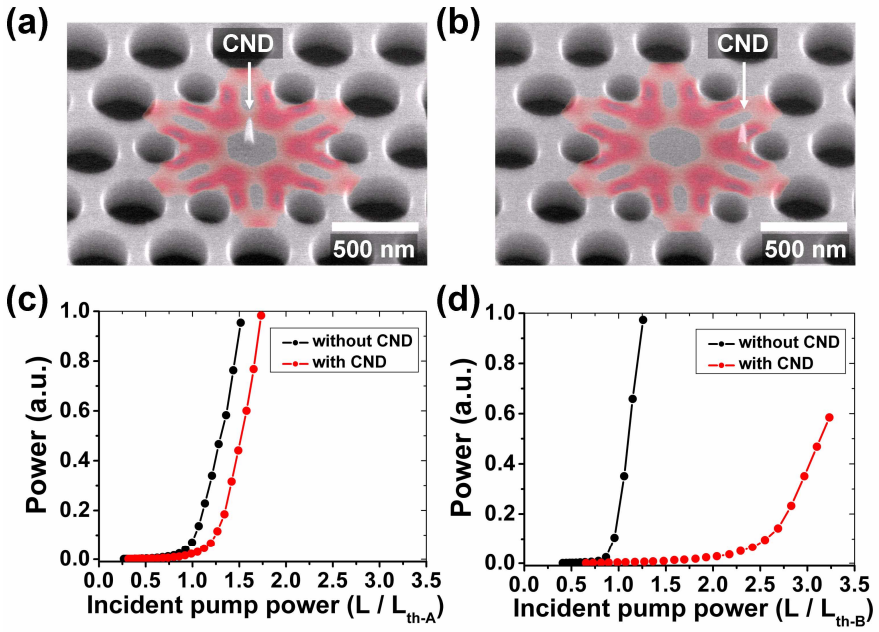


Fig. 7. Effects of CND on lasing characteristics. (a) Deposition of a CND at the center, where the hexapole mode has a node. (b) Deposition of a CND at the antinode of the hexapole mode. Overlaid is the photon density profile of the hexapole mode. (c) (d) Lasing characteristics modified by the CND placed at the node(c) and antinode(d), respectively.

6. Optical characteristics of carbonaceous nano-dots

Despite the small size of <100 nm, the CND causes quite a large resonance shift. In fact, such a dramatic resonance shift cannot be observed by deposition of dielectric material with size of <100 nm. This means that the CND constructed by EBID has partially metallic characteristics. To understand the optical characteristics of the CNDs, we perform FDTD simulations by introducing a nano-dot with a diameter of ~50 nm in the intensity antinode of the hexapole mode of a PhC single-cell cavity. The simulation shows that the dielectric nano-dots with refractive indices varying from 1.5 to 3.0 cause a spectral red-shift less than 1 nm. This result is apart from the measured tuning value of >2 nm, which is generated by the CND with a diameter of ~50 nm in experiment. On the other hand, if we consider that the CND is made of a metal with an optical constant similar to that of gold or silver in the FDTD calculations, the measured large tuning can be well understood. Consequently, the CND has metallic characteristics causing absorption and scattering but does not spoil the high Q factors of the resonant modes too much. Therefore, the resonant spectrum can effectively be modified.

As further applications, EBID gives full play to its ability when various metallic or dielectric materials such as gold, silver, and silicon dioxide are deposited [24, 25]. Especially, EBID silicon oxide nano-structures without strong absorption and scattering would be useful to achieve more precise and careful tuning of a PhC nano resonator. In addition, one can construct a thin EBID film on a bare or patterned surface by scanning on a certain area by the focused electron beam. The thin film and patterned nanostructures consisting of metallic or dielectric EBID materials can be used for manifold optical applications such as tuning of a specific area, generation of efficient nonlinear signals, and excitation of the surface plasmons [29].

7. Summary

We successfully demonstrated sub-nanometer spectral tuning of a high-Q PhC laser mode by electron beam induced CND deposition. A CND with a diameter of <100 nm was fabricated

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in a desired position using scanning electron microscope. Precise and selective spectral tuning was observed with a resolution of <0.2 nm, comparable to the linewidth of the high-Q resonant mode. The nano-dot of controllable size and position alters lasing characteristics depending on the electric field profile of the mode. We observed that the lasing threshold of the PhC mode is affected by a nano-dot placed in the strong electric field region. We believe that these results show a meaningful step toward the functionalization of optical nanocavities for practical single photon sources and photonic integrated circuits.

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